8051 Trainer Kit - 12 V DC Motor Experiment

Complete first-year wiring, circuit diagram and operating procedure

Purpose and boundary

This experiment teaches basic 8051 interfacing using START, STOP, RESET, a simulated FAULT, TCRT5000 RPM pulses, LEDs, buzzer and an FR120N motor switch. It is a separate teaching experiment and does not replace the complete ESP32/MQTT condition-monitoring system.

1. Aim

To interface a small 12 V geared DC motor with an AT89S52/AT89C52-compatible trainer kit and demonstrate fail-safe motor OFF at boot, manual START, latched STOP/fault shutdown, separate RESET, RPM counting and visual/audible status indication.

2. Trainer-kit requirements

Requirement	Expected value	Why it matters
MCU	AT89S52, AT89C52 or standard compatible 8051	Code uses REG51.H and standard port/interrupt registers.
Crystal	11.0592 MHz	Produces 9600-baud UART using Timer 1. Other crystals require TH1 recalculation.
Logic supply	Regulated +5 V	8051, TCRT5000, LEDs and buzzer logic.
Exposed ports	P1, P2 and P3 headers	Required to reach the assigned pins.
Programmer	Kit ISP/USB programmer or compatible external programmer	Loads the generated HEX file.

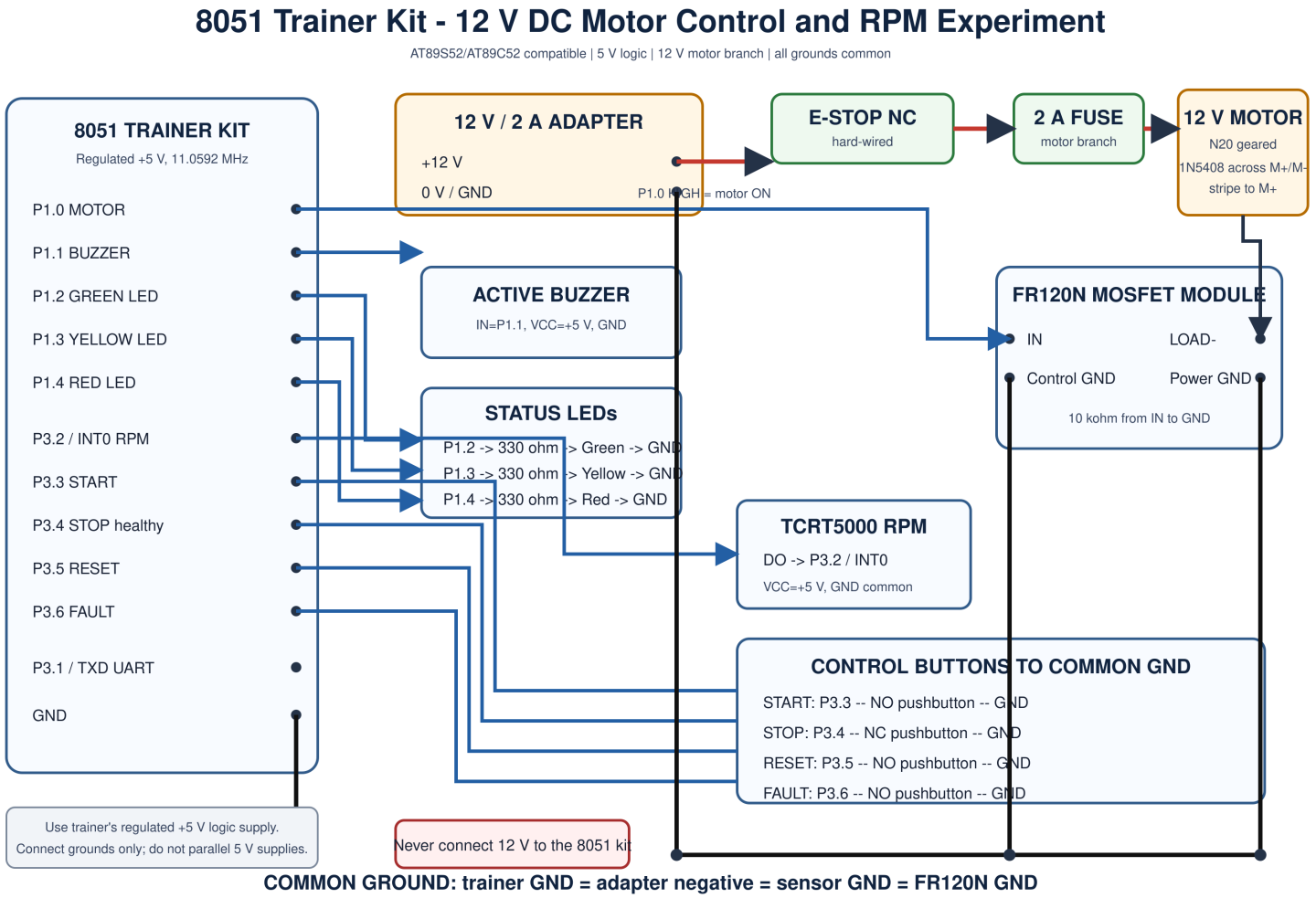
3. Minimum experiment components

Only low-voltage parts required for the first-year trainer-kit experiment

Component	Qty	Required specification	Use
8051 trainer kit	1	AT89S52/AT89C52, +5 V	Controller and programming platform
12 V N20 geared motor	1	Approx. 300 RPM, below 1 A normal current	Controlled load
FR120N MOSFET module	1	5 V control compatible	Low-side motor ON/OFF
12 V adapter	1	Isolated, 2 A	Motor power
E-stop NC + fuse	1 each	Latching NC; 2 A fuse	Hard-wired motor cutoff
TCRT5000 module	1	Digital output	One pulse/revolution
Buttons	4	START NO, STOP NC, RESET NO, FAULT NO	User control and safe fault simulation
LEDs + resistors	3 + 3	Green/yellow/red; 330 ohm	RUN/READY/TRIP status
Active buzzer module	1	5 V logic compatible	Trip indication
Protection parts	1 set	1N5408, 100 nF, 470 uF/25 V, 10 kohm	Flyback, noise reduction and safe gate state

4. Complete circuit diagram

Use the pin labels printed on the trainer kit. Connector position varies between manufacturers.



5. 8051 pin map and exact connections

Critical rule

The trainer uses regulated 5 V logic. The motor uses the separate 12 V adapter. Connect grounds together, but never connect 12 V to an 8051 trainer pin. The 8051 output controls only the FR120N input; it never drives the motor directly.

8051 pin	Connection	Active state	Exact external wiring
P1.0	FR120N IN	HIGH = motor ON	P1.0 to IN; 10 kohm from IN to GND; common control GND
P1.1	Active buzzer IN	HIGH = ON	Module VCC to +5 V and GND to common ground
P1.2	Green RUN LED	HIGH = ON	P1.2 -> 330 ohm -> anode; cathode -> GND
P1.3	Yellow READY LED	HIGH = ON	P1.3 -> 330 ohm -> anode; cathode -> GND
P1.4	Red TRIP LED	HIGH = ON	P1.4 -> 330 ohm -> anode; cathode -> GND
P3.2 / INT0	TCRT5000 DO	Falling-edge pulse	Sensor VCC +5 V, GND common, DO to P3.2
P3.3	START NO	LOW = pressed	Normally-open button between P3.3 and GND
P3.4	STOP NC	LOW = healthy	Normally-closed contact between P3.4 and GND; HIGH means STOP/open wire
P3.5	RESET NO	LOW = pressed	Normally-open button between P3.5 and GND
P3.6	FAULT NO	LOW = fault	Normally-open button between P3.6 and GND
P3.1 / TXD	UART output	9600 baud	Use trainer onboard serial interface or a correctly level-shifted receiver

Motor power branch

12 V wiring

12 V adapter + -> latching emergency-stop NC -> 2 A fuse -> motor +
Motor - -> FR120N switched LOAD- terminal -> adapter negative
1N5408 across motor: striped cathode at motor +, anode at motor -

5 V logic wiring

Use the trainer's normal regulated 5 V supply. Join trainer GND, FR120N GND, TCRT5000 GND and adapter negative. Do not parallel two 5 V supplies unless the trainer manual explicitly allows it.

6. Working and program behavior

Action or condition	Motor	Indicator	Trip state
Power ON	OFF	Yellow READY	Clear
Press START while safe	ON	Green RUN	Clear
Press STOP or open STOP wire	OFF immediately	Red + buzzer	Latched
Press FAULT	OFF immediately	Red + buzzer	Latched
Press START after a trip	Remains OFF	Red + buzzer	Still latched
Restore safe inputs and press RESET	Remains OFF	Yellow READY	Cleared
Press START after RESET	ON	Green RUN	Clear

Program flow

```
BOOT -> MOTOR OFF -> initialize UART and INT0
|
+-> STOP open or FAULT active? -> latch trip -> MOTOR OFF + red LED + buzzer
|
+-> START edge and safe? -> MOTOR ON + green LED
|
+-> RESET edge and safe? -> clear trip; MOTOR remains OFF
|
+-> every 1 second -> RPM = counted INT0 pulses x 60 -> UART
```

Build in Keil uVision

Step	Procedure
1	Create a new project and select the trainer's exact MCU, normally Atmel AT89S52.
2	Add 8051/src/main.c to Source Group 1 and confirm REG51.H is available.
3	Set the target crystal to 11.0592 MHz and enable Create HEX File.
4	Build, program the HEX using the trainer programmer and reset the kit.
5	Open the trainer's UART terminal at 9600 baud, 8 data bits, no parity, 1 stop bit.

Code location

GitHub: https://github.com/MayankGangwar1234/motor_monitoring/tree/main/8051 - source file: 8051/src/main.c

7. Optional full-sensor expansion

These connections preserve the existing project hardware, but their driver code should be taught as later experiments. They are not required for the simple START/STOP/RPM program.

Device	8051 connection	Supply	Mandatory note
INA219	SDA via level shifter to P2.0; SCL via level shifter to P2.1	3.3 V logic; VIN+/VIN- in 12 V branch	I2C address 0x40; never pull sensor bus up to 5 V
MPU6050	Shared level-shifted I2C bus on P2.0/P2.1	3.3 V	Address 0x68; rigid motor mounting
SSD1306 OLED	Shared level-shifted I2C bus on P2.0/P2.1	3.3 V	Address normally 0x3C
DS18B20	DATA to P1.5	5 V	4.7 kohm pull-up from DATA to +5 V
TCRT5000	DO to P3.2 / INT0	5 V	Adjust for one clean falling edge per revolution

I2C level-shifter connection

```
8051 +5 V -> BSS138 module HV LV <- regulated 3.3 V
8051 P2.0 -> HV1 LV1 -> INA219/MPU6050/OLED SDA
8051 P2.1 -> HV2 LV2 -> INA219/MPU6050/OLED SCL
Common GND --------------------------------> all sensor grounds
```

8. Precautions and expected result

- Keep the complete experiment low-voltage and supervised.
- Test P1.0 and the FR120N input with the motor disconnected before the first run.
- Verify STOP reads LOW when healthy. A HIGH STOP input must force a trip.
- Never stall or hold the motor shaft to create a fault; use the FAULT button.
- Fix the motor and reflective RPM disc firmly and keep loose wires away.
- If the trainer's onboard LEDs or buzzer are active-LOW, use external parts as drawn or invert the code outputs.

Expected result

The 8051 starts the motor only from a safe READY state, stops and latches on STOP/fault, requires a separate RESET followed by START, counts one RPM pulse per revolution and reports RPM through UART.

Reference documents

Microchip AT89S52 datasheet; Vishay TCRT5000 datasheet; purchased trainer-kit schematic/manual; purchased FR120N module terminal labels. The exact trainer and MOSFET module labels take priority over connector position.